

COMP 3270 – Introduction to Algorithms

Homework 7

Topics – Heap sorts

1. Illustrate the operation of MAX-HEAPIFY (A, 1) on the array A = {27, 17, 3, 16, 13, 10, 1, 5, 7, 12, 4, 8, 9, 0}.
2. Illustrate the operation of HEAPSORT on the array A = {5, 13, 2, 25, 7, 17, 20, 8, 4}.
3. For HEAPSORT codes below,

```
Heapsort(A)
{
    Build-MAX-Heap(A);
    for (i = A.length downto 2)
    {
        Swap(A[1], A[i]);
        A.heap_size = A.heap_size - 1;
        MAX-Heapify(A, 1);
    }
}
```

(a) What is the number of required swap operations when heapsort the array A = {5, 13, 2, 25, 7, 17, 20, 8, 4}? Explain your reason.

(b) If we replace MAX-Heapify(A, 1) with Build-MAX-Heap(A), what is the number of required swap operations when heapsort the array A? Explain your reason